A Simple Structure of LLC Resonant DC-DC Converter for Multi-output Applications

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***Abstract*—Multi-output LLC resonant topologies with asymmetrical structures dedicating to multi-output power supply systems, having limitation of lowest load, are presented in this paper. According to output power level, different configuration of secondary output end can be flexibly adapted. Consequently, the presented topology is a favorable candidate for such application as whose outputs are not symmetrical, i.e. compared with auxiliary output, power level of main output is comparatively higher. In addition, all switches of the very topology operate under the condition of ZVS and therefore it is applicable to such occasion as high frequency, high power density. Asymmetrical operating principle and ZVS operation are analyzed in detail, moreover key parameter design considerations are presented. Finally, the analysis above is verified both by simulation and experiment.**

Keywords-resonant; multi-output; LLC

I. INTRODUCTION

In the power supply system, various loads often require various types of voltage value. Both cost and complexity of the very power supply system can be reduced once the multi-output topology is adopted. Even though, some types of power products, which have no requirement of multi-output, whose controlling IC and driving IC require several auxiliary power supplies, therefore it is necessary to supply multi-outputs for the products themselves.

Flyback converter is the preferable topology in multi-output occasion due to the simplicity of its secondary configuration, wherein one winding, one diode and one capacitor are included in one output unit. However, as is well known, flyback converter is merely applicable to such application as whose power level is comparatively low. Forward, half-bridge, and push-pull are appropriate for application occasion of high power, however compared with the other topologies, the secondary configurations of the above mentioned topologies are further complex, and that costs increase correspondingly. For the application, whose main output power is extremely high however auxiliary output is comparatively low, the above listed converters are not suitable. LLC resonant type converter possesses many merits and is widely adopted in application of high power density and high efficiency. Both bridge-mode rectifier and full-wave rectifier circuit with center-tapped transformer can be applied in the secondary configuration of the very topology and filtering inductor L is not required, therefore the configuration complexity is accordingly reduced. However, for the application of ultra-low power, the above secondary output configuration is still somewhat complex, for bridge-mode rectifier circuit involves 4 diodes and center-tapped rectifier circuit involves 2 diodes and 2 windings.

For some specific application, for example net power supply, communication power supply and so on, requirement of lowest load current limitation of main output exists and full range load current is needless. An LLC resonant type DC/DC converter with asymmetrical multi-output configuration, which dedicates to such application as that requirement of lowest load current limitation exists, is presented in this paper. The output circuit configuration on secondary side can be flexibly adapted accordingly to output power level, therefore the very topology is appropriate to the application whose main output power is high, whereas whose auxiliary output power is ultra low. In addition, all semiconductor devices involved in this topology operate under the condition of soft switching and that comparatively high switching frequency can be applied, correspondingly size of the power supply can be greatly reduced.

II. SECONDARY CIRCUIT CONFIGURATION OF LLC RESONANT CONVERTER

Both full bridge circuit and half bridge circuit can be usually adopted on primary side of LLC resonant converter, and that the former configuration is the preferable candidate provided that power level is extremely high, whereas once power level is comparative low, the later configuration is an alternative. Usually, as shown in Fig.1 (a) and (b), either full-bridge current type full-wave rectifier or current type full-wave rectifier with center-tapped transformer can be adopted on secondary side, the former rectifier circuit is preferable if load is comparatively heavy, whereas once output voltage is extremely high, the later one is favorable. Current type half-wave rectifier circuit, whose configuration is especially simple, as shown in Fig.1 (c), is usually adopted in flyback topology. The very configuration is not recommended in LLC resonant converter with single output, however in application of multi-output, it can be applied on secondary side of auxiliary output. The very output configuration can draw energy from primary side merely during half of every operation period and consequently results in asymmetry of current waveform of the other outputs, therefore multi-output LLC resonant converter involving the third secondary output configuration can be named asymmetrical multi-output LLC resonant converter.

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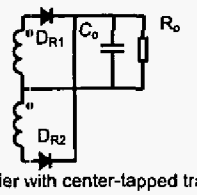
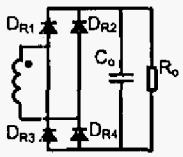
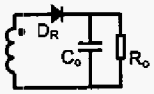


(a)full-wave rectifier with center-tapped transformer



(b) full-bridge full-wave rectifier



(c) half-wave rectifier

Figure 1. Secondary side structures for LLC resonant converter

III. Operation Principle Analysis

Here, operation principle of single output LLC series resonant DC/DC converter is not specified, referring to reference [3~8]. A dual-output LLC resonant converter, whose outputs consist of current-mode full-bridge full-wave rectifier circuit and current-mode half-wave rectifier circuit, is shown in Fig.2. In the following text, operation principle of asymmetrical multi-output LLC resonant converter is specified based on the above presented structure of multi-output LLC resonant converter. Output, whose configuration is current-mode full-bridge full-wave rectifier, is supposed to be main, correspondingly, the other output, whose configuration is current-mode half-wave rectifier, is supposed to be auxiliary and that closed loop control strategy is adopted on main output. In the following analysis, it is assumed that both output capacitors are infinite and that constant voltage sources of V_{o1} and V_{o2} can be used to substitute of them. MOSFET reversely paralleled with parasitic diode is used as main switch. One switching period of the very converter can be divided into 8 operation stages, whose operation waveforms and equivalent circuit of each stage are respectively shown in Fig.3 and Fig.4. Each operation stage is specified as follows:

(1) Stage 1:t0~t1. At the moment of t0, S_1 is turned on, primary resonant current circulates through S_1 and gradually increases in sinusoidal mode. On main output side, D_{R1} and D_{R4} conduct and correspondingly, secondary side of transformer is clamped by voltage of main output, therefore resonance occurs between L_s and C_s on the primary side and that magnetizing current of i_{Lm} circulating through L_m increases linearly, whereas on auxiliary output side, D_{R5} is reverse blocking. Therefore, current of main output i_{o1} is proportional to the difference of resonant current and magnetizing current on the primary side, which is specifically illustrated in Fig.3. Once i_p and i_{Lm} equals to each other, the very operation stage terminates.

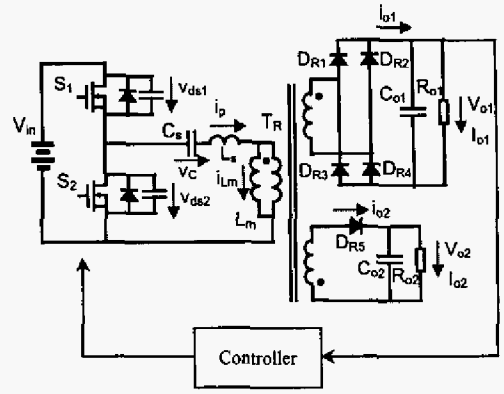


Figure 2. Asymmetrical structure multi-output LLC resonant converter

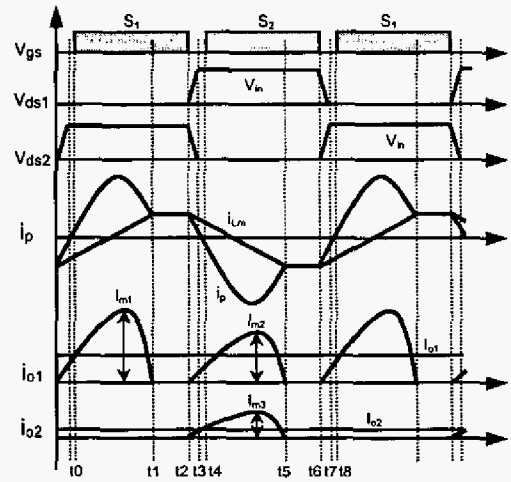


Figure 3. Principle waveform of the proposed converter

(2) Stage 2: t1~t2. At the moment of t1, main output current i_{o1} decreases to zero due to the equivalency of i_p and i_{Lm}, consequently, D_{R1} and D_{R4} are turned off under the condition of ZCS and no reverse recovery occurs, because current circulating through them decreases to zero naturally. Meanwhile, clamping effect of main output disappears, therefore L_m acts as resonant inductor and resonance occurs among L_m, L_s and C_s. Compared with the above resonant period, the present one is greatly longer, therefore, current of primary side can be regarded as constant during this interval and L_m is supposed to be a constant current source. Once driving signal of S_1 disappears, this stage terminates.

(3) Stage 3: t2~t3. At the moment of t2, S_1 is turned off, therefore current of primary side i_p charges to the parasitic capacitor of S_1, simultaneously discharges to the parasitic capacitor of S_2, and at the same time D_{R2} and D_{R3} conduct on the main output side, secondary side of transformer is clamped by main output voltage again, correspondingly D_{R5} conduct on the auxiliary output end. Once parasitic capacitor

of S_2 is discharged to zero, this interval terminates.

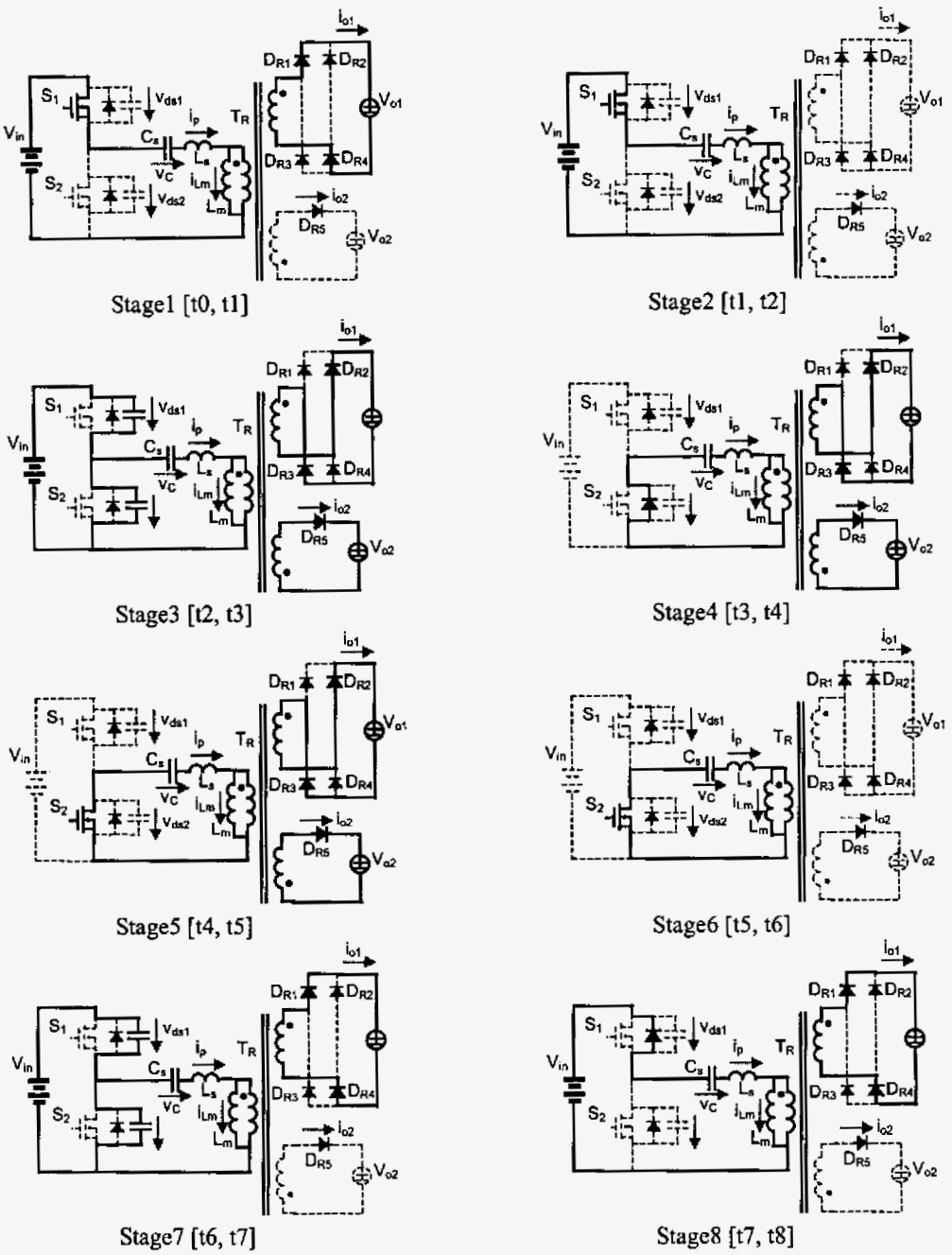


Figure 4. Equivalent circuit for each stage

(4) Stage 4: t3~t4. When voltage across parasitic capacitor of S_2 decreases to zero, current on primary side begins to circulate through body diode of S_2 and that reversely increases in sinusoidal mode. On main output side, D_{R2} and D_{R3} are turned on and on auxiliary output side, D_{R5} is turned on. Voltage across drain to source of S_2 remains zero due to conduction of body diode of S_2 and ZVS of S_2 is easily achieved. Once driving signal of S_2 appears, the very stage terminates.

(5) Stage 5:t4~t5. At the moment of t4, S_2 is turned on, primary resonant current is carried by S_2 and gradually increases to reverse direction as sinusoid. On main output side, D_{R2} and D_{R3} conduct and correspondingly, secondary side of transformer is still clamped by voltage of main output, therefore resonance occurs between L_s and C_s on the primary

side and that magnetizing current i_{Lm} circulating through L_m decreases linearly, whereas on auxiliary output side, D_{R5} conduct. Therefore, summation of current of main output i_{o1} and current of auxiliary output i_{o2} on the secondary side is proportional to the difference of resonant current and magnetizing current on the primary side, which is specifically illustrated in Fig.3. Once i_p and i_{Lm} equals to each other, the very operation stage is over.

(6) Stage 6: t5~t6. At the moment of t5, main output current i_{o1} decreases to zero due to the equivalency of i_p and i_{Lm}, consequently, D_{R2} and D_{R3} are turned off under the condition of ZCS and no reverse recovery occurs, because current circulating through them decreases to zero naturally. Meanwhile, clamping effect of main output disappears, therefore L_m acts as resonant inductor and resonance occurs among L_m, L_s and C_s. Certainly, D_{R5} is turned off under the condition of ZCS as well. Once driving signal of S_2 disappears, this stage terminates.

(7) Stage 7: t6~t7. At the moment of t6, S_2 is turned off, therefore current of primary side i_p charges to the parasitic capacitor of S_2, simultaneously discharges to the parasitic capacitor of S_1., and at the same time D_{R1} and D_{R4} conduct on the main output side, secondary side of transformer is clamped by main output voltage again, correspondingly D_{R5} is reverse blocking on the auxiliary output end. Once voltage across parasitic capacitor of S_1 is discharged to zero, this interval terminates.

(8)Stage 8: t7~t8. When voltage across parasitic capacitor of S_1 decreases to zero, current on primary side begins to circulate through body diode of S_1 and that positively increases as sinusoid. Voltage across drain to source of S_1 remains zero due to conduction of body diode of S_1 and ZVS of S_1 is easily achieved. Once driving signal of S_1 appears, the very stage terminates and another switching period begins.

According to the above analysis, energy of the auxiliary output end can be drawn from primary side during half of every switching period, and for the other half period it is supplied by output capacitor. For main output, during one switching period, current waveforms of each half period are not exactly the same, which results in difference of current stress of D_{R1}, D_{R4} and D_{R2}, D_{R3}. However, operation detail of primary side is exactly the same as that of symmetrical LLC resonant type converter.

IV. DESIGN CONSIDERATION

Operation process of primary side circuit of asymmetrical multi-output LLC resonant type converter is exactly the same as that of ordinary LLC resonant type converter, therefore their design consideration that is specified in references is similar. In the following text, design consideration of secondary side circuit involved in asymmetrical multi-output LLC resonant converter is specified.

A. REQUIREMENT OF LOAD

Due to existence of asymmetry on the secondary side, some specific requirement of load must be satisfied to guarantee the normal operation of primary side. It is assumed that a typical asymmetrical multi-output LLC resonant converter includes three types of auxiliary output configurations, as shown in Fig.1, on the secondary side. Wherein, number of output circuits consisting of full-wave rectifier configuration with center-tapped transformer, as shown in Fig.1 (a), is n1, i.e.V_{oa1}, V_{oa2}, …, V_{oan1}, number of output circuits consisting of bridge-mode full-wave rectifier configuration, as shown in Fig.1 (b), is n2, i.e. V_{ob1}, V_{ob2}, …, V_{obn2}. According to the difference of relative end of the transformer, half-wave rectifier, which is shown in Fig.1 (c), can be divided into two categories, i.e. positive and negative. Provided that, number of positive ones is n3, i.e. V_{oc1}, V_{oc2}, …, V_{ocn3}, number of negative ones is n4, i.e. V_{od1}, V_{od2}, …, V_{odn4}. Then the following expression must be satisfied:

$$\left|\sum_{i=1}^{n3} V_{oci} - \sum_{i=1}^{n4} V_{odi}\right| \le \sum_{i=1}^{n1} V_{oai} + \sum_{i=1}^{n2} V_{obi} \tag{1}$$

According to expression (1), conclusion that output power of main side mustn't less than that of auxiliary side can be obtained for the proposed topology, which is shown in Fig.2.

B. SELECTION OF DIODE

Selection of diode on secondary side is important for asymmetrical multi-output configurations. Reverse voltage flat is the only factor of voltage stress consideration for diodes on secondary side because ZCS condition for which are achieved and no voltage spike occurs. For the output configurations, as shown in Fig.1 (a) and (c), voltage stress of diodes is twice of output voltage, whereas for the output configuration, as shown in Fig.1 (b), voltage stress of diode is exactly the same as output voltage. Analysis of current stress of diode is comparatively complex and is specified based on the circuit, as shown in Fig.2, as follows.

Resonant period of Ls and Cs is defined as Ts and can be expressed as follows:

$$T_s = 2\pi\sqrt{L_s C_s} \tag{2}$$

The interval of t5-t2 is half of Ts. Actually, current waveform circulating through secondary diode of the very interval is the difference of a half sinusoid current and magnetizing current and it is supposed to be a half sinusoid for the convenience of engineering calculation.

As shown in Fig.3, I_{m1}, I_{m2} and I_{m3} are the respective peak value of three half sinusoidal waveforms and the following expressions can be obtained:

$$\frac{I_{m1}T_s}{\pi T} + \frac{I_{m2}T_s}{\pi T} = I_{o1} \tag{3}$$

$$\frac{I_{m3}T_s}{\pi T} = I_{o2} \tag{4}$$

$$\frac{I_{m1}T_s V_{o1}}{\pi T} = \frac{I_{m2}T_s V_{o1}}{\pi T} + \frac{I_{m3}T_s V_{o2}}{\pi T} \tag{5}$$

Wherein T is the switching period.

The following expressions can be concluded according to expression (3), (4) and (5):

$$I_{m1} = \frac{\pi T I_{o1}}{2T_s} + \frac{\pi T I_{o2} V_{o2}}{2T_s V_{o1}} \quad (6)$$

$$I_{m2} = \frac{\pi T I_{o1}}{2T_s} - \frac{\pi T I_{o2} V_{o2}}{2T_s V_{o1}} \quad (7)$$

$$I_{m3} = \frac{\pi T I_{o2}}{T_s} \quad (8)$$

I_{m1}, I_{m2}, I_{m3}, whose correspondingly average value are I_1, I_2, I_3, are the respective peak value of D_{R1} and D_{R4}, D_{R2} and D_{R3}, D_{R5}.

$$I_1 = \frac{I_{o1}}{2} + \frac{I_{o2} V_{o2}}{2V_{o1}} \quad (9)$$

$$I_2 = \frac{I_{o1}}{2} - \frac{I_{o2} V_{o2}}{2V_{o1}} \quad (10)$$

$$I_3 = I_{o2} \quad (11)$$

V. SIMULATION AND EXPERIMENTAL RESULTS

For verifying the correctness of the above analysis, both simulation and experiment results are given. Circuit structure, as shown in Fig.2, is adopted in both simulation and experiment, whose specification and main parameter are as follows.

Input voltage V_{in}: AC 220V±20% (after full-bridge rectifier circuit); Main output voltage V_{o1}: 48V; Main output current I_{o1}: 1~2A; Auxiliary output voltage V_{o2}: 24V; Auxiliary output current I_{o1}: 0~0.5A; Switching frequency of normal voltage and full load f: 100kHz; Main switches S_1, S_2: IRF830; Rectifier diodes of main output D_{R1}, D_{R2}, D_{R3}, D_{R4}: 10CPQ60; Rectifier diodes of auxiliary output D_{R5}: SR506; Transformer T: n=64:18:9, L_m=750uH, Leak inductor : 70uH; Resonant inductor L_s: 80uH; Resonant capacitor C_s: 13.2nF.

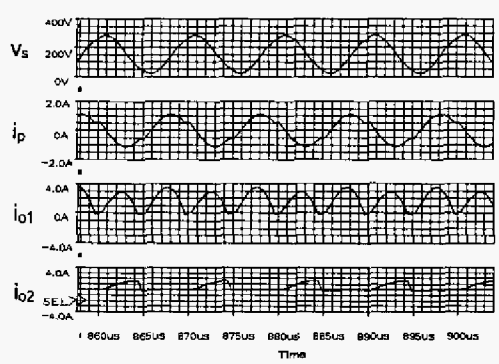
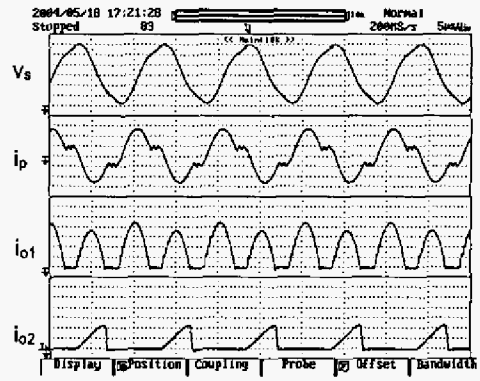


Figure 5. Simulation waveform

Fig.5 shows the simulation waveforms of normal input voltage and full load, which was obtained by PSPICE, and correspondingly, Fig.6 (a) shows the experimental waveforms. It is obviously indicated that both simulation and experimental results meet the above principle analysis. Fig.6 (b) shows the voltage waveforms across GS and DS of S_1, it is illustrated that voltage across GS begins to increase after voltage across DS decreasing to zero, which either match the above principle analysis. The circumstance with S_2 is as well. Therefore, both S_1 and S_2 are turned on under the condition of ZVS. Efficiency of prototype under normal input and at full load reaches 93%.



V_s: 50V/div; i_p: 0.5A/div; i_{o1}: 1A/div; i_{o2}: 1A/div
time: 5us/div

(a) Resonant waveform

V_{gs}: 5V/div; V_{ds}: 100V/div; time: 500ns/

(b) Voltage waveforms across GS and DS of S_1

Figure 6. Experimental waveform

VI. CONCLUSION

Asymmetrical multi-output LLC resonant converter is the preferable topology for such application as whose main output power is relatively high and auxiliary output power is further low, which attributes to its characteristic of high performance and low cost.

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